

Intrinsic Projector Realization

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Summary

Pass 75 replaces the explicit support and stage tags from the tagged restricted pro-Ab realization by internal projectors.

The new target is still a certificate target, not yet an LCA-sheaf or condensed realization. The progress is that support and stage data are no longer external labels.

Projectors

Support is encoded by commuting Boolean idempotents e_p :

$$e_S = \prod_{p \in S} e_p, \quad e_S e_T = e_{S \cap T}.$$

The lcm tower stage is encoded by projectors q_n :

$$q_n q_m = q_{\min(n, m)}.$$

The realization

$$\rho_{\text{proj}} : \mathcal{H}_\epsilon \rightarrow \mathcal{R}_\epsilon^{\text{proj}}$$

sends finite conductor windows, Loeb-Rosser boundaries, restrictions, signed duals, and lcm stages to finite/pro abelian data equipped with these projector actions.

Result

The projector-enriched realization is faithful on the five generator families in the checked finite/pro window.

The plain target that forgets projector actions is still not faithful. Thus the support and stage structure are mathematically necessary.

A companion no-go checker is also integrated: ordinary exact 1-category realization is insufficient because the $\widehat{\mathbb{Z}}/\mathbb{Z}$ term is recovered by $R^1 \varprojlim$, not by a finite exact cone.

Verification

The main checker `code/scripts/check-pass75.py` produces `artifacts/reports/pass75-intrinsic-projector-realization-check.json`.

It verifies:

- 75 projector-enriched generators through six primes, $k \leq 3$, and N_{24} ;
- zero collisions for projector signatures;
- 12 collisions in the plain target;
- Boolean support-projector relations;
- 576 stage-projector relations;
- restriction source/target recovery by projector action.

The integrated companion checker `code/scripts/check-pass73-exact-obstruction.py` produces `artifacts/reports/pass73-exact-realization-obstruction-check.json` and records the ordinary exact-category obstruction.

The next task is to realize e_p and q_n naturally in an established target: for example as support projectors on a finite-prime stratified site and a derived pro-stage filtration carrying $\varprojlim^1$.